

RotorLab — Steady MRF Results Report

Results and discussion for the steady MRF benchmark run. Full geometry, mesh, and physics setup detail lives in the companion Build & Decision Log — this document covers the converged result and what it means.

1. Objective

Benchmark a standard Rushton turbine stirred tank ($T = 0.300$ m, $D = 0.100$ m, $N = 5$ rev/s = 300 rpm, water) using steady MRF in Simcenter STAR-CCM+, and compare the resulting Power Number against the published value of $Po \approx 5.0$ for this geometry.

2. Method Summary

788,728-cell trimmed mesh across two regions (RotatingFluid 604,488 cells, StationaryFluid 184,240 cells) joined by an Internal/In-place/Imprinted interface. Realizable K-Epsilon Two-Layer turbulence, Two-Layer All Y+ wall treatment, segregated flow, water at 998 kg/m^3 / $0.001 \text{ Pa}\cdot\text{s}$. RotatingFluid solved in a rotating reference frame at 31.416 rad/s (Motion: Stationary, Reference Frame: Rotating — the mesh does not move; only the frame the equations are solved in does). Full detail in the Build & Decision Log.

3. Convergence

Run to Maximum Steps = 1500 (the Asymptotic torque criterion was disabled partway through the run after it repeatedly stopped the solver on a loose 2%/50-sample band while the torque was still undergoing slow monotonic drift — see Section 4). Final residuals:

Residual	Final value (iter. 1500)	Behavior
Continuity	2.58×10^{-4}	~3.6 orders of magnitude drop from start; small periodic oscillation after iter. ~1300
X-momentum	1.81×10^{-3}	steady decline throughout
Y-momentum	1.00×10^{-3}	steady decline, tightest of the momentum residuals
Z-momentum	4.92×10^{-3}	slowest to converge (~9x drop from its iter. 184 plateau of 0.09), but genuinely still improving through 1500, not stalled
Tke	1.81×10^{-3}	steady decline
Tdr	8.53×10^{-3}	persistent oscillation between $\sim 4.6 \times 10^{-3}$ and 1.2×10^{-2} from iter. ~700 onward; likely a practical floor for this quantity in this class of flow, not non-convergence

Overall: 2-3+ orders of magnitude reduction across all monitored residuals, consistent with acceptable engineering convergence for a rotating-machinery RANS case. Continuity and Y-momentum reached the tightest levels; Tdr's persistent low-amplitude oscillation is the only residual that didn't fully settle, which is common in MRF cases given the inherently unsteady flow features (trailing vortices) a steady frame can only represent in an iteration-averaged sense.

4. Torque and Power Number

Moment 1 (torque on ImpellerWall about the rotation axis) converged to **-0.41666 N·m** — stable to 5 significant figures across the final 10 iterations (range: -0.416664 to -0.416665 N·m). The negative sign is a direction convention, not a physical issue.

Quantity	Value
Converged torque magnitude	0.41666 N·m
$Po = \text{Torque} \times 2\pi N / (\rho N^3 D^5)$	0.41666×25.18
Computed Power Number (this run)	$Po \approx 10.49$
Published benchmark	$Po \approx 5.0$
Ratio (this run / benchmark)	$\approx 2.1\times$

5. Near-Wall Mesh Resolution Check

Wall Y^+ on ImpellerWall ranges 0.127 to 63 (target was 20-30). The colored contour, however, shows the entire visible impeller surface — disc, blades, shaft — as uniformly low Y^+ (near the 0.127 end), meaning the wetted area is predominantly over-resolved relative to target, not under-resolved. The $Y^+ = 63$ maximum is confined to a small, localized region not visible as a distinct patch at this viewing angle (most likely a blade tip or edge singularity), rather than a widespread resolution problem. This weakens mesh resolution as the primary explanation for the Po gap in Section 6 — a small, localized patch contributes little to an integrated torque over the whole surface.

6. Discussion — Why $Po \approx 10.5$ Instead of ≈ 5.0

Three candidate explanations were checked, in order of how easy they'd be to fix:

Setup error (ruled out). Rotation rate (31.416 rad/s = 300 rpm, confirmed directly in the Motion object) and the Moment report's Axis/Origin ([0,0,1]/[0,0,0], matching the Motion exactly) were both verified correct.

Mesh resolution (unlikely to be the primary driver). Per Section 5, the elevated Y^+ region is small and localized, not widespread — unlikely to account for a factor of ~ 2 on its own.

Blade-baffle interaction frozen out by steady MRF (most likely primary driver). Both fluid regions are solved together — StationaryFluid (containing the four wall baffles) is fully coupled to RotatingFluid across the interface, so this is not a case of only simulating a small rotating zone. The gap instead comes from what steady MRF cannot represent anywhere in that coupled domain: the Rushton's radial discharge jet fires directly at the baffles, and in reality each blade sweeps past each baffle in turn, producing a strong, genuinely periodic blade-baffle interaction. Steady MRF solves one frozen angular position — the equivalent of a single strobe-lit snapshot — so that sweep, and the periodic torque signal it drives, cannot develop. Realizable K-Epsilon is further documented to over-predict power draw for this geometry class on top of that. A $\sim 2\times$ over-prediction from this method-and-model combination, on a baffled Rushton benchmark specifically, is consistent with known behavior, not evidence of a modeling mistake. Sliding mesh is the direct test of this explanation: it lets the mesh physically rotate so the blade-baffle sweep actually happens.

7. Next Steps

Build and run the transient sliding-mesh case on the identical geometry and mesh. Unlike MRF, sliding mesh physically rotates the mesh through multiple impeller revolutions, allowing the blade-passage unsteadiness this steady result cannot capture to actually develop. Comparing MRF, sliding mesh, and the $Po \approx 5.0$ benchmark against each other is the core result this project is built around.